LINEAR AND ROBUST CONTROL SYSTEMS

ELEC 9731

7 - Tracking and Disturbance Rejection

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Topics

1. 1 DOF Review
2. 2 DOF issues:
2 DOF fails in tracking and need 3 DOF design.
3. State Space View
4. Transfer Function View

NB. Error space method of tracking design assumes full state observed.

Notation. DR=Disturbance Rejection.

FR=filtered reference.

References Goodwin et al chapters 7,17,18

1 DOF Review

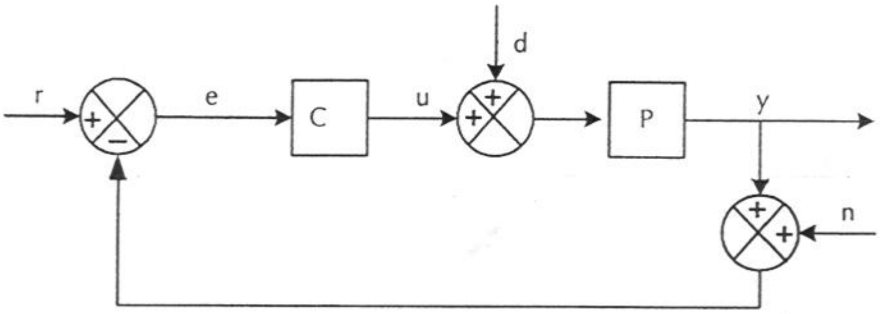

FIGURE 7.1: Standard 1DOF Feedback Control

- Closed loop Laplace Transforms are

$$\begin{aligned}\overline{y} &= \overline{y}_r + \overline{y}_d + \overline{y}_n \\ &= SPC\overline{r} + SP\overline{d} - SPC\overline{n} \\ \mathcal{S} &= \frac{1}{1+PC}\end{aligned}$$

- Also tracking error signal LT is

$$\begin{aligned}\overline{e} &= \overline{r} - \overline{y} \\ &= \overline{e}_r + \overline{e}_d + \overline{e}_n \\ &= \mathcal{S}\overline{r} - SP\overline{d} + SPC\overline{n}\end{aligned}$$

- Control Signal

$$\begin{aligned}\overline{u} &= \overline{u}_r + \overline{u}_d + \overline{u}_n \\ &= \mathcal{S}C\overline{r} - \mathcal{S}PC\overline{d} - \mathcal{S}C\overline{n}\end{aligned}$$

KEY TRANSFER FUNCTIONS

•For tracking and disturbance rejection, four closed loop transfer functions matter

•Tracking Error TF

$$G_{r\to e}(s) = \mathcal{S}(s) = \frac{1}{1+P(s)C(s)} = \frac{aa_c}{a_{cL}}$$

• Tracking Control TF

$$G_{r\to u}(s) = \mathcal{S}(s)C(s) = \frac{aa_c}{a_{cL}}\frac{b_c}{a_c} = \frac{ab_c}{a_{cL}}$$

•Disturbance Rejection TF

$$G_{d\to y}(s) = \mathcal{S}(s)P(s) = \frac{ba_c}{a_{cL}}$$

• Disturbance Control TF

$$G_{d\to u}(s) = -T(s) = -\frac{bb_c}{a_{cL}}$$

•Design ensures a_{cL} is stable.

TRACKING - 1DOF

•Model the reference signal as

$$\overline{r}(s) = \frac{\beta_r(s)}{\alpha_r(s)}$$

e.g. step $\frac{1}{s}$, ramp $\frac{1}{s^2}$

step + sinusoid $\frac{1}{s} + \frac{a}{s^2+w_0^2} = \frac{s^2+w_0^2+as}{s(s^2+w_0^2)}$

• Steady State Tracking Error is

$$\begin{aligned} e_{ss,r} &= \lim_{s\to 0} s\overline{e}_r(s) \\ &= \lim_{s\to 0} sG_{r\to e}(s)\overline{r}(s) \\ &= \lim_{s\to 0} s.\frac{aa_c}{a_{cL}}\frac{\beta_r}{\alpha_r} \end{aligned}$$

This gives 0 if

$\alpha_r(s)$ is a factor of $a_c(s)$

i.e. if Internal Model Principle holds

i.e. if Controller contains a model of reference (denominator)

NB. It may happen that a cancels part of α_r but that would be fortuitous; we cannot assume it.

Tracking Example

DC-motor + PI control

•DC-motor model ($\tau_{mech} >> \tau_{elec} = \tau$)

$$\overline{\theta}(s) = \frac{B}{s(\tau s + 1)}\overline{v}_a(s) = \frac{b(s)}{a(s)}\overline{v}_a$$

Controller (PI)

$$\begin{aligned} C(s) &= K + \frac{K_I}{s} = K + \frac{1}{T_I s} \\ &= \frac{Ks + K_I}{s} = \frac{b_c(s)}{a_c(s)} \end{aligned}$$

Tracking Example - Continued

- First check stability

$$
\begin{aligned}
a_{cL} &= bb_c + aa_c \\
&= B(Ks + K_I) + s(\tau s + 1)s \\
&= \tau s^3 + s^2 + BKs + BK_I
\end{aligned}
$$

Routh stability criterion requires

$$a_1 > 0, a_3 > 0, a_1 a_2 > a_3$$

$$a_1 = 1 > 0$$

$$a_3 = BK_I > 0 \Rightarrow K_I > 0$$

$$a_1 a_2 > a_3 \Rightarrow BK > BK_I \Rightarrow K > K_I$$

- Check IMP

a_c has s as a factor, so closed loop system can track a step.

NB. Here it happens that $a(s)$ has a factor s so actually a ramp can be tracked.

TRACKING CONTROL - 1DOF

Steady State control signal value is

$$\begin{aligned} u_{ss,r} &= \lim_{s\to 0} s\overline{u}_r(s) \\ &= \lim_{s\to 0} sG_{r\to u}(s)\overline{r}(s) \\ &= \lim_{s\to 0} s\frac{ab_c}{a_{cL}}\frac{\beta_r}{\alpha_r} \end{aligned}$$

This is bounded if $\alpha_r(s)|sb_c(s)a(s)$

This limits capacity of IMP since b_c, a_c are coprime.

• Step: $\Rightarrow \alpha_r(s) = s$ so need

$s|a_c(s)a(s)$ & $s|sb_c(s)a(s)$

i.e. $s|a_c(s)$ & $b_c(s)$ free $\Rightarrow$ OK.

• Ramp: $\Rightarrow \alpha_r(s) = s^2$ so need

$s^2|a_c(s)a(s)$ & $s^2|sb_c(s)a(s)$

i.e. $s^2|a_c(s)a(s)$ & $s|b_c(s)a(s)$

So cannot track a ramp in general unless it happens that $s|a(s)$.

DISTURBANCE REJECTION - 1DOF

•Disturbance Model

$$\overline{d}(s) = \frac{\beta_d(s)}{\alpha_d(s)}$$

•Steady State Output is

$$\begin{aligned} y_{ss,d} &= \lim_{s\to 0} s\overline{y}_d(s) \\ &= \lim_{s\to 0} sG_{d\to y}(s)\overline{d}(s) \\ &= \lim_{s\to 0} s\frac{ba_c}{a_{cL}}\frac{\beta_d}{\alpha_d} \end{aligned}$$

•This gives 0 if $\alpha_d(s)$ is a factor of $a_c(s)$

i.e. if IMP holds for controller

•Example - DC motor + PI continued

Since s is a factor of a_c a disturbance step can be rejected.

DISTURBANCE CONTROL - 1DOF

Steady state control is

$$
\begin{aligned}
u_{ss,d} &= \lim_{s\to 0} s\overline{u}_d \\
&= \lim_{s\to 0} sG_{d\to u}(s)\overline{d}(s) \\
&= \lim_{s\to 0} s\frac{-bb_c}{a_{cL}}\frac{\beta_d}{\alpha_d}
\end{aligned}
$$

which is bounded if $\alpha_d | sb_c$.

Again this limits range of IMP.

- Step: $\Rightarrow \alpha_d(s) = s$ so need

$s|a_c(s)b(s)$ & $s|sb_c(s)b(s)$

i.e. $s|a_c(s)$ & $b_c(s)$ free $\Rightarrow$ OK.

- Ramp: $\Rightarrow \alpha_d(s) = s^2$ so need

$s^2|a_c(s)b(s)$ & $s^2|sb_c(s)b(s)$

i.e. $s^2|a_c(s)b(s)$ & $s|b_c(s)b(s)$

So cannot reject a ramp in general unless it happens that $s|b(s)$.

example plots for 1dof dist rej/tracking

2 DOF Review (LSFBO)

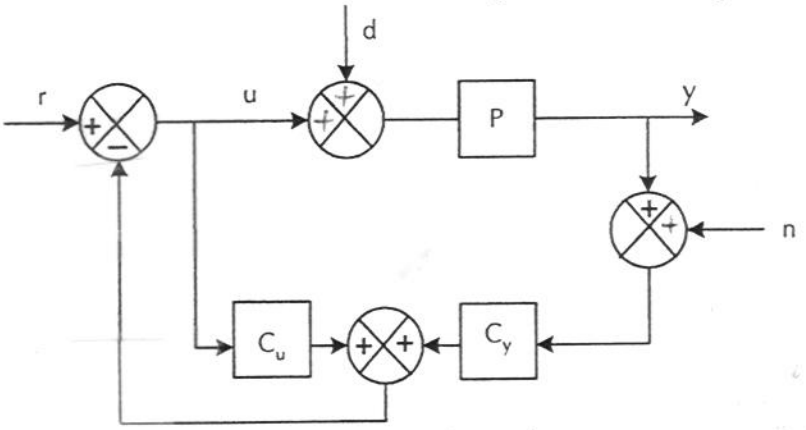

FIGURE 7.2: LSFBO 2DOF architecture

- Closed loop equations are

$$
\begin{aligned}
\overline{y} &= \overline{y}_r + \overline{y}_d + \overline{y}_n \\
&= \frac{P}{D}\overline{r} + \frac{(1+C_u)P}{D}\overline{d} - \frac{PC_y}{D}\overline{n} \\
&= \frac{P}{D}\overline{r} + SP\overline{d} - T\overline{n} \\
\mathcal{S} &= \frac{1+C_u}{D} = \frac{1+C_u}{1+C_u+PC_y} \\
T &= 1-\mathcal{S} = \frac{PC_y}{D}
\end{aligned}
$$

- Control TFs

$$
\begin{aligned}
\overline{u} &= \overline{u}_r + \overline{u}_d + \overline{u}_n \\
&= \frac{\overline{r}}{D} - \frac{PC_y}{D}\overline{d} - \frac{C_y}{D}\overline{n}
\end{aligned}
$$

2 DOF - Tracking and Disturbance Rejection

- Again only four TFs matter
- Tracking Error TF

$$G_{r \to e} = 1 - \frac{P}{D} = 1 - G = 1 - \frac{b}{a_k}$$

- Tracking Control TF

$$G_{r \to u} = \frac{1}{D} = \frac{1}{1 + C_u + PC_y} = \frac{a}{a_k}$$

- Disturbance Rejection TF (IDTF)

$$G_{d \to y} = \mathcal{S}P = \frac{b(a_l + b_u)}{a_k a_l} = G + GC_u$$

- Disturbance Control TF

$$G_{d \to u} = \frac{-PC_y}{D} = -T = NTF = -\frac{bb_y}{a_l a_k} = -GC_y$$

Note the Disturbance related TFs show that C_u, C_y have direct external control importance.

2DOF - TRACKING (LSFBO)

- Reference Signal Model

$$\overline{r}(s) = \frac{\beta_r(s)}{\alpha_r(s)}$$

- Steady State Tracking Error

$$\begin{aligned} e_{ss,r} &= \lim_{s \to 0} s\overline{e}_r(s) \\ &= \lim_{s \to 0} sG_{r \to e}(s)\overline{r}(s) \\ &= \lim_{s \to 0} s\left[1 - \frac{b(s)}{a_k(s)}\right]\frac{\beta_r(s)}{\alpha_r(s)} \end{aligned}$$

- To get 0, we need $\alpha_r(s)$ to be a factor of $a_k(s) - b(s)$ but $a_k(s)$ is determined by closed loop stability and performance whereas $b(s)$ is fixed. So no ability to ensure $\alpha_r(s)$ is a factor. Similarly for tracking control.

- So new design needed.

2 DOF - Disturbance Rejection

- Disturbance Model

$$\bar{d}(s) = \frac{\beta_d(s)}{\alpha_d(s)}$$

- Steady State Output

$$\begin{aligned} y_{ss,d} &= \lim_{s\to 0} sG_{d\to y}(s)\bar{d}(s) \\ &= \lim_{s\to 0} s\frac{b(a_l + b_u)}{a_k a_l}\frac{\beta_d}{\alpha_d} \end{aligned}$$

- To get 0 we need

$b(a_l + b_u)$ to contain α_d as a factor. But b is fixed and a_l is determined by observer stability while b_u is determined through the Bezout design equation from a_k, a_l No ability to ensure α_d is a factor. Similarly for disturbance control.

- So design change needed.

2 DOF TRACKING DESIGN

- Design Modification

\- is to prefilter reference signal

- System

$$\dot{x} = Ax + bu$$

- Observer + LSFB

$$\dot{\hat{x}} = A\hat{x} + bu + l(y - c^T\hat{x}) - mr$$

The gain vector m is to be chosen

- Control

$$u = -k^T\hat{x} + m_0 r$$

[m_0 is a scale factor to be chosen]

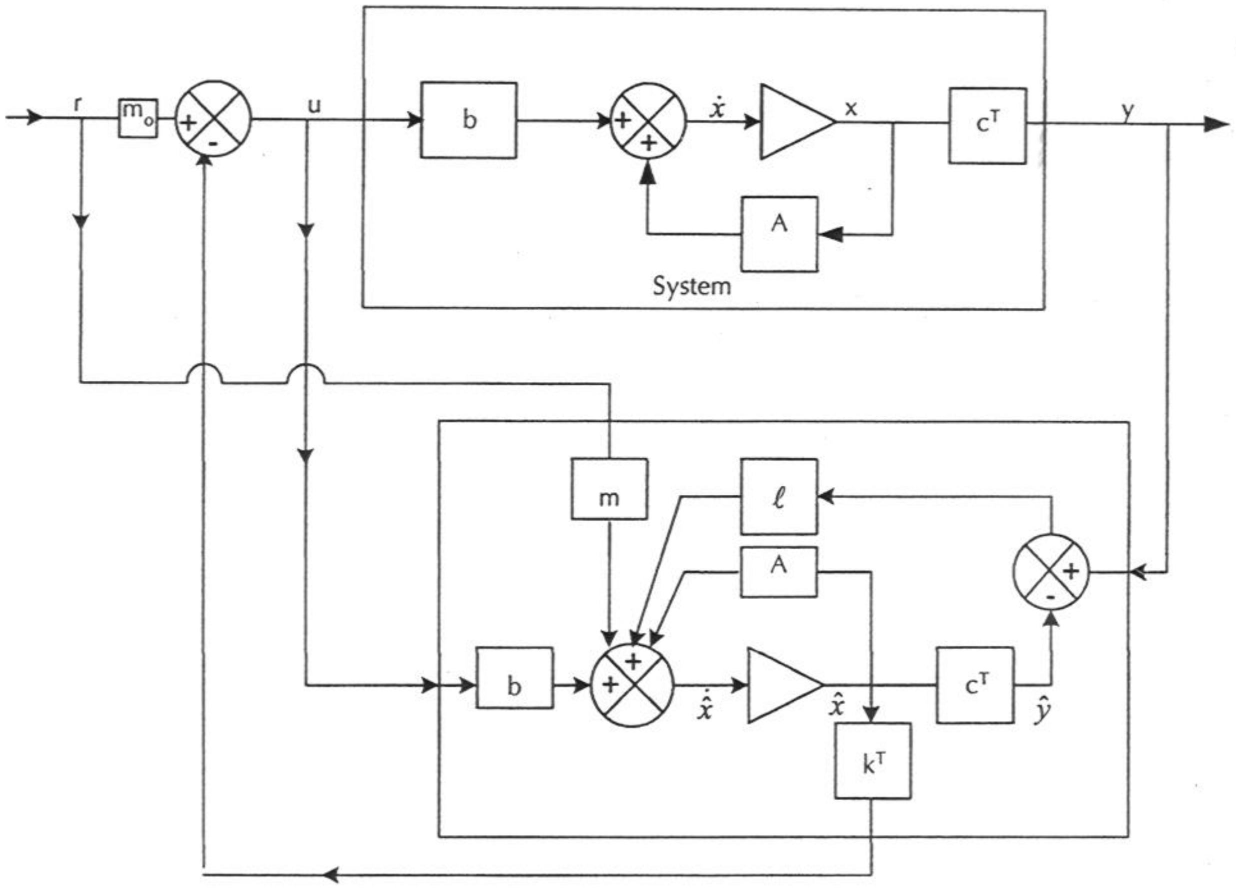

Fig. 7.3: LSFBO + filtered reference for tracking

LSFB+0+FR ANALYSIS

- As usual form the error state equation ($\tilde{x} = x - \hat{x}$) to find

$$\dot{\tilde{x}} = (A - lc^T)\tilde{x} + mr$$

- Also rewrite system as follows

$$\begin{aligned} \dot{x} &= Ax + b(m_0 r - k^T \hat{x}) \\ &= (A - bk^T)x + bk^T \tilde{x} + bm_0 r \end{aligned}$$

- Joint SS equations are then

$$\begin{pmatrix} \dot{x} \\ \dot{\tilde{x}} \end{pmatrix} = \begin{pmatrix} A - bk^T & +bk^T \\ 0 & A - lc^T \end{pmatrix} \begin{pmatrix} x \\ \tilde{x} \end{pmatrix} + \begin{pmatrix} bm_0 \\ m \end{pmatrix} r$$

TRACKING ANALYSIS

Take Laplace transforms and invert to find

$$\begin{pmatrix} sI - A + bk^T & -bk^T \\ 0 & sI - Alc^T \end{pmatrix} \begin{pmatrix} \overline{x} \\ \overline{\tilde{x}} \end{pmatrix}$$

$$= \begin{pmatrix} bm_0 \\ m \end{pmatrix} \overline{r} + IC$$

$$\Rightarrow \begin{pmatrix} \overline{x} \\ \overline{\tilde{x}} \end{pmatrix}$$

$$= \begin{pmatrix} []_k^{-1} & []_k^{-1} bk^T []_l^{-1} \\ 0 & []_l^{-1} \end{pmatrix} \begin{pmatrix} bm_0 \\ m \end{pmatrix} \overline{r} + IC$$

$$[]_k = sI - A + bk^T$$

$$[]_l = sI - A + lc^T$$

TRACKING ANALYSIS - Continued

$$
\begin{aligned}
\Rightarrow \overline{x} &= \left\{[]_k^{-1} b m_0 + []_k^{-1} b k^T []_l^{-1} m\right\} \overline{r} + IC \\
\Rightarrow \overline{y} &= c^T \overline{x} \\
&= c^T []_k^{-1} b(m_0 + k^T []_l^{-1} m)\overline{r} + IC
\end{aligned}
$$

Recall $c^T []_k^{-1} b = \frac{b}{a_k}$

Define $b_r = k^T adj(sI - A + lc^T)^{-1} m$

$$\Rightarrow \overline{y} = \frac{b}{a_k} \frac{(m_0 a_l + b_r)}{a_l} \overline{r} + IC$$

So the CLTF has been altered. And a new set of zeros introduced that can be shaped.

TRANSFER FUNCTION VIEW

As usual return to observer equation to exhibit TF aspects

$$
\begin{aligned}
\dot{\hat{x}} &= A\hat{x} + bu + l(y - c^T\hat{x}) - mr \\
&= (A - lc^T)\hat{x} + bu + ly - mr
\end{aligned}
$$

Take Laplace transforms and invert

$$
\begin{aligned}
\Rightarrow \overline{\hat{x}} &= (sI - A + lc^T)^{-1} b\overline{u} \\
+(sI - A + lc^T)^{-1} l\overline{y} &- (sI - A + lc^T)^{-1} m\overline{r} \\
\Rightarrow \overline{u} &= m_0\overline{r} - k^T\overline{\hat{x}} \\
&= -C_u\overline{u} - C_y\overline{y} + C_r\overline{r}
\end{aligned}
$$

TRANSFER FUNCTION VIEW - Continued

Where

$$
\begin{aligned}
C_u &= k^T(sI - A + lc^T)^{-1}b \\
&= \frac{b_u}{a_l} \\
C_y &= k^T(sI - A + lc^T)^{-1}l \\
&= \frac{b_y}{a_l} \\
C_r &= k^T(sI - A + lc^T)^{-1}m + m_0 \\
&= \frac{b_r + m_0 a_l}{a_l} \\
&= \frac{B_r}{a_l}
\end{aligned}
$$

This gives a (constrained) 3 DOF design as in Fig7.4

3 DOF view of LSFBO + FR

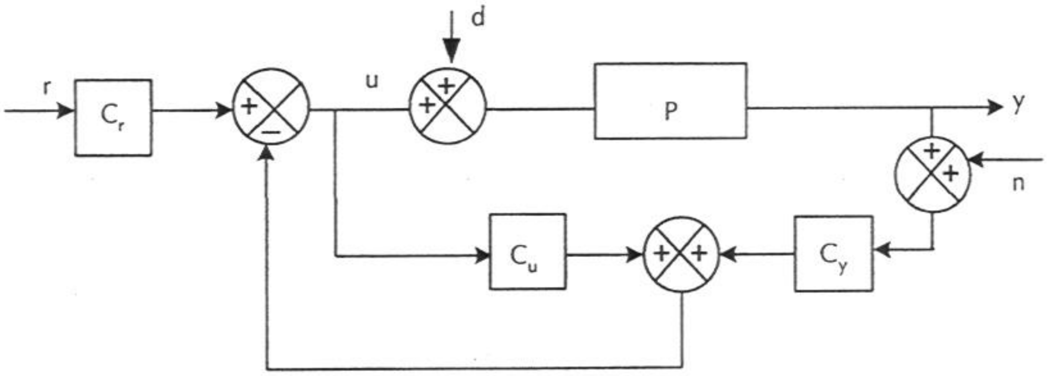

FIGURE 7.4: 3 DOF architecture for LSFBO with reference signal tracking

3 DOF TF ANALYSIS

•The closed loop equations are as before but $C_r\overline{r}$ replaces $\overline{r}$

$$\begin{aligned}
\overline{y} &= \overline{y}_r + \overline{y}_d + \overline{y}_n \\
&= \frac{PC_r}{D}\overline{r} + \mathcal{S}P\overline{d} - T\overline{n} \\
\overline{u} &= \overline{u}_r + \overline{u}_d + \overline{u}_n \\
&= \frac{C_r}{D}\overline{r} + \frac{PC_y}{D}\overline{d} - \frac{C_y}{D}\overline{n}
\end{aligned}$$

3 DOF TF TRACKING ANALYSIS - Continued

- The tracking error TF is

$$\begin{aligned}
G_{r\to e} &= 1 - \frac{PC_r}{D} \\
&= 1 - \frac{\frac{b}{a}\frac{(m_0 a_l + b_r)}{a_l}}{1 + \frac{b_u}{a_l} + \frac{b}{a}\frac{b_y}{a_l}} \\
&= 1 - \frac{bB_r}{a(a_l + b_u) + bb_y}, B_r = (b_r + m_0 a_l) \\
&= 1 - \frac{bB_r}{a_k a_l} \\
&= \frac{a_k a_l - bB_r}{a_k a_l}
\end{aligned}$$

- Tracking Control TF

$$G_{r\to u} = \frac{C_r}{D} = \frac{aB_r}{a_k a_l}$$

NB Must check that $\mathcal{S} = \frac{dG/G}{dP/P}$ is unchanged.

3 DOF TF Tracking Analysis - Continued

•The steady state error is

$$
\begin{aligned}
e_{ss,r} &= \lim_{s \to 0} s\bar{e}_r(s) \\
&= \lim_{s \to 0} sG_{r \to e}(s) \frac{\beta_r(s)}{\alpha_r(s)} \\
&= \lim_{s \to 0} s \frac{[a_k a_l - bB_r]}{a_k a_l} \frac{\beta_r}{\alpha_r}
\end{aligned}
$$

• This is 0 if

$\alpha_r(s)$ is a factor of $a_k a_l - bB_r$

3 DOF TF Tracking Analysis - Continued III

i.e. we need

$$a_k a_l - bB_r = \alpha_r \phi$$

where ϕ is a polynomial of suitable degree

$$\begin{aligned} \Rightarrow a_k a_l &= \alpha_r \phi + bB_r \\ &= \alpha_r \phi + b[m_0 a_l + b_r] \end{aligned}$$

If we set $m_0 = 1$ we get

$$\begin{aligned} a_k a_l &= \alpha_r \phi + b(a_l + b_r) \\ \Rightarrow a_l(a_k - b) &= \alpha_r \phi + bb_r \end{aligned}$$

If we set $m_0 = 0$ we get

$$a_k a_l = \alpha_r \phi + bb_r$$

- And so long as b, α_r are coprime.

We can solve this tracking Bezout equation to get b_r, ϕ [for any m_0]

TRACKING PARAMETER COUNTS

- Control Design Bezout Equation

$$a_k a_l = a(a_l + b_u) + bb_y$$

Check that highest powers match

monic(n_a)$\times$ monic(n_a) = monic(n_a) x monic(n_a) + lower degree terms $\Rightarrow$ OK.

Check equations and unknowns

$$\begin{array}{rcl} 2n_a\text{equations} & \leftrightarrow & \#b_u + \#b_y \\ & = & n_a(\text{since degree} = n_a - 1) \\ & + & n_a(\text{since degree} = n_a - 1) \end{array}$$

which match $\Rightarrow$ OK.

TRACKING BEZOUT EQUATION

•Tracking Bezout Equation (with $m_0 = 0$)

$$a_k a_l = \alpha_r \phi + b b_r$$

Check highest powers

monic(n_a)× monic(n_a)= monic(n_r)×monic(n_ϕ) + lower degree terms

$\Rightarrow \phi$ must be monic of degree $n_\phi = 2n_a - n_r$

Check equations and unknowns

$$\begin{array}{rcl} 2n_a\text{equations} & \leftrightarrow & \#\phi + \#b_r \\ & = & 2n_a - n_r + n_r(\text{to match total}) \end{array}$$

TRACKING BEZOUT EQUATION
Continued

•But b_r could have degree up to $n_a - 1$; and need deg $b_r <$ deg a_l

So how to match with n_r unknowns?

Simplest but still general way is to put

$$\begin{aligned}
b_r &= b_{or}\psi \\
\deg(b_{or}) &= n_r - 1 \text{with} n_r \text{coefficients}) \\
\deg(\psi) &= n_a - n_r \\
\psi & \quad \text{is monic and chosen separately with} \\
& \quad n_a - n_r - 1 \text{ coefficients}
\end{aligned}$$

Bezout equation becomes

$$a_k a_l = \alpha_r \phi + b b_{or} \psi$$

•How to choose ψ?

Recall CLTF (with $m_0 = 0$)

$$G_{r \to y} = \frac{PC_r}{D} = \frac{bb_r}{a_k a_l} = \frac{bb_{or}\psi}{a_k a_l}$$

So choose ψ to shape additional closed loop zeros.

TRACKING CONTROL

We have

$$\begin{aligned}
u_{ss,r} &= \lim_{s\to 0} sG_{r\to u}\bar{r} \\
&= \lim_{s\to 0} s\frac{ab_r}{a_k a_l}\frac{\beta_r}{\alpha_r} \\
&= \lim_{s\to 0} s\frac{ab_{or}\psi}{a_k a_l}\frac{\beta_r}{\alpha_r}
\end{aligned}$$

We consider $\lim_{s\to 0}\frac{sb_{or}\psi}{\alpha_r}$ for special cases of α_r.

(i) Step. OK.

(ii) Sinusoid/Step+Sinusoid. OK.

(iii) Ramp. Not possible in general.

TRACKING BEZOUT EQUATION
Continued II

•Simpler alternative is to take ψ to be a factor of a_l so write

$$\begin{aligned} a_l &= a_{or}\psi, a_{or}\text{monic of deg} = n_r \\ \Rightarrow G_{r\rightarrow y} &= \frac{b}{a_k}\frac{b_{or}}{a_{or}} \end{aligned}$$

Note then that

$$C_r = \frac{b_r}{a_l} = \frac{b_{or}\psi}{a_{or}\psi} = \frac{b_{or}}{a_{or}}$$

This simplifies the Bezout equation as follows:
Put $\phi = a_\phi \psi$ where a_ϕ is monic of degree n_a

$$\begin{aligned} \Rightarrow a_k a_l &= \alpha_r \phi + b b_{or}\psi \\ \Rightarrow a_k a_{or}\psi &= \alpha_r a_\phi \psi + b b_{or}\psi \\ \Rightarrow a_k a_{or} &= \alpha_r a_\phi + b b_{or} \end{aligned}$$

TRACKING BEZOUT EQUATION
Continued III

Check highest powers of this equation

monic(n_a)$\times$monic(n_r)
=monic(n_r)$\times$monic(n_a)+low degree terms

$\Rightarrow$O.K.

Check # equations and parameters

$n_a + n_r$ equations $\leftrightarrow \#a_\phi + \#b_{or}$

$= n_a + n_r \Rightarrow$O.K.

•If we go back to the original requirement that α_r divides $a_k a_l - bB_r$ we find

$$\begin{aligned} &= a_k a_l - bb_r, \ (m_0 = 0) \\ &= a_k a_{or}\psi - bb_{or}\psi \\ &= (a_k a_{or} - bb_{or})\psi \end{aligned}$$

Since ψ is fixed by design we need

α_r divides $a_k a_{or} - bb_{or}$

So put $a_k a_{or} - bb_{or} = \alpha_r a_\phi$ which agrees with discussion above.

3DOF TRACKING State Space View

This simplified process is equivalent to processing r through its own observer instead of through the observer for x.

We use a SS model for r

$$\dot{\hat{x}}_r = A_r x_r, r = c_r^T x_r$$

which leads to the 2DOF design

$$\dot{\hat{x}} = A\widehat{x} + bu + l(y - c^T \hat{x})$$

together with

$$\dot{\hat{x}} = A_r \hat{x}_r + l_r(r - c_r^T \hat{x}_r)$$

and a control law

$$u = -k^T \hat{x} - k_r^T \hat{x}_r$$

State Space View - Continued

Taking Laplace transforms we find

$$\begin{aligned}\overline{u} &= -k^T(sI - A + lc^T)^{-1}b\overline{u} \\ -k^T(sI - A + lc^T)^{-1}l\overline{y} &- k_r^T(sI - A_r + l_r c_r^T)^{-1}l_r\overline{r} \\ &= -C_u\overline{u} - C_y\overline{y} - C_r\overline{r}\end{aligned}$$

With C_u, C_y as before and where now

$$C_r = \frac{b_{or}}{a_{or}}, a_{or} = \alpha_r + l_{or}$$

The rest of the analysis now follows as before.

Tracking Design Summary

(i) Specify $a_k, a_l = a_{or}\psi$ where

a_k(degree n_a)

a_{or}(degree n_r=degree of α_r)

ψ (degree $n_a - n_r$) ;are monic, stable

(ii) Solve observer controller Bezout equation

$a_k a_l = a(a_l + b_u) + bb_y$for b_u, b_y

(iii) Solve tracking Bezout equation

$a_k a_{or} = \alpha_r a_\phi + bb_{or}$ for a_ϕ, b_{or}

(iv) Implement 3 DOF design getting

$C_u = \frac{b_u}{a_l}, C_y = \frac{b_y}{a_l}, C_r = \frac{b_{or}}{a_{0r}}$

NB. Since $a_l = a_{or}\psi$ must do two jobs i.e. estimate states of x and x_r there may be tradeoffs in this approach. The other approach with ψ separately chosen may sometimes be more attractive.

3dof tracking example

3DOF Disturbance Rejection Design

• Idea is to estimate d from a dynamic model of it and subtract off the estimate

• SS System

$$\dot{x} = Ax + b(u + d), \; y = c^T x$$

•Disturbance Model

$$\dot{x}_d = A_d x_d, \; d = y_d = c_d^T x_d$$

NB: $\overline{d} = c_d^T (sI - A)^{-1} x_{d0} = \frac{\beta_d(s)}{\alpha_d(s)}$

Disturbance Rejection Design - Continued

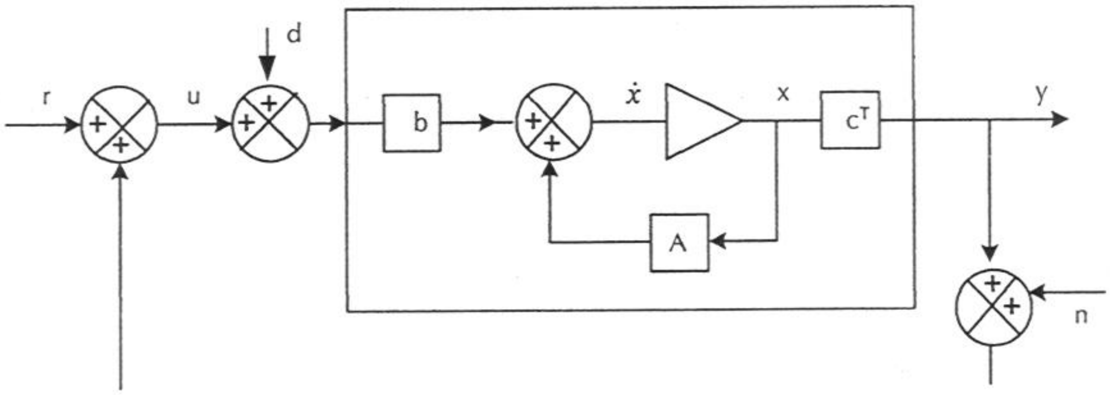

FIGURE 7.5: Disturbances in a SS model

3DOF DISTURBANCE REJECTION DESIGN

•Joint Model

$$\begin{aligned}
\dot{x}_J &= A_J x_J + b_J u,\ y = c_J^T x_J \\
A_J &= \begin{pmatrix} A_d & 0 \\ b c_d^T & A \end{pmatrix}, b_J = \begin{pmatrix} 0 \\ b \end{pmatrix}, x_J = \begin{pmatrix} x_d \\ x \end{pmatrix} \\
c_J^T &= \begin{pmatrix} 0^T & c^T \end{pmatrix}
\end{aligned}$$

• Model is observable (check) but not controllable. Assume A, b, c controllable.

•Observer

$$\dot{\hat{x}}_J = A_J \hat{x}_J + b_J u + l_J (y - c_J^T \hat{x}_J)$$

3DOF DISTURBANCE REJECTION DESIGN - Continued

- Control

$$
\begin{aligned}
u &= -k^T\hat{x} + mr - \hat{d} \\
&= -k^T\hat{x} - c_d^T\hat{x}_d + mr \\
&= -(c_d^T, k^T)\hat{x}_J + mr \\
&= -k_J^T\hat{x}_J + mr
\end{aligned}
$$

$\Rightarrow$'big' LSFBO problem

- Partition

$$
k_J = \begin{pmatrix} c_d \\ k \end{pmatrix}, l_J = \begin{pmatrix} l_d \\ l \end{pmatrix}
$$

DISTURBANCE REJECTION DESIGN Continued III

•Aim to express in terms of parameters of disturbance free design and check IMP.

•We need to calculate

$b_{Ju}, b_{Jy}, a_{Jk}, a_{Jl}$

• CLAIMS

$$\begin{aligned} a_{Jl} &= \alpha_d a_l + b l_d \\ a_{Jk} &= a_k \\ a_{Jl} + b_{Ju} &= \alpha_d (a_l + b_u) \\ b_{Jy} &= l_d a_k + b_y \alpha_d \end{aligned}$$

•Then recompute $y_{ss,d}$

3DOF IMP

- Disturbance rejection TF

$$
\begin{aligned}
G_{d\rightarrow y} &= \mathcal{S}_J P_J \\
&= \frac{b}{a}\frac{\left(1+\frac{b_{Ju}}{a_{Jl}}\right)}{\left(1+\frac{b_{Ju}}{a_{Jl}}+\frac{b}{a}\frac{b_{Ju}}{a_{Jl}}\right)} \\
&= \frac{b(a_{Jl}+b_{Ju})}{a_{Jk}a_{Jl}} \\
&= \frac{b\alpha_d(a_l+b_u)}{a_k a_{Jl}}
\end{aligned}
$$

- IMP

$$
\begin{aligned}
y_{ss,d} &= \lim_{s\rightarrow 0} sG_{d\rightarrow y}(s)\frac{\beta_d}{\alpha_d} \\
&= \lim_{s\rightarrow 0} s\frac{b\alpha_d(a_l+b_u)}{a_k a_{Jl}}\frac{\beta_d}{\alpha_d}
\end{aligned}
$$

Now we have cancellation of α_d, so $y_{ss,d} = 0$. And so disturbance rejection is established.

IMP - Continued

- Disturbance Control TF

$$
\begin{aligned}
G_{d\to u} &= \frac{-PC_y}{D} \\
&= -\frac{bb_{Jy}}{a_{Jl}a_{Jk}} \\
&= -\frac{b(l_d a_k + b_y \alpha_d)}{(\alpha_d a_l + b l_d) a_k} \\
&= -\frac{b(l_d a_k + b_y \alpha_d)}{a_{Jl} a_k}
\end{aligned}
$$

Now consider $\lim_{s\to 0} s\frac{b(l_d a_k + b_y \alpha_d)}{\alpha_d}$ for special cases of α_d

(i) Step. OK.

(ii) Sinusoid/Step+Sinusoid. OK.

(iii) Ramp. Not possible in general.

3DOF DR Design Summary

(i) Specify a_k, a_{Jl}

(ii) Solve Joint Bezout equation

$a(a_{Jl} + b_{Ju}) + bb_{Jy} = a_k a_{Jl}$ for b_{Ju}, b_{Jy}

(iii) Implement design using

$C_{Ju} = \frac{b_{Ju}}{a_{Jl}}, C_{Jy} = \frac{b_{Jy}}{a_{Jl}}$

However using relations established above we can simplify the design.

REDUCED DR DESIGN

Substitute into the Bezout equation to find

$$\begin{aligned} & a\alpha_d(a_l + b_u) + b(l_d a_k + b_y \alpha_d) \\ = \quad & a_k(\alpha_d a_l + b l_d) \\ \Rightarrow \quad & a\alpha_d(a_l + b_u) + b b_y \alpha_d = a_k \alpha_d \alpha_l \end{aligned}$$

Cancelling α_d divisor gives

$$a(a_l + b_u) + b b_y = a_k a_l$$

i.e. LSFBO Bezout equation

REDUCED DR DESIGN - Continued

(i) Specify a_{Jl}, a_k

(ii) Solve Bezout equation

$a_{Jl} = \alpha_d a_l + b l_d$ for a_l, l_d

Solution exists provided b, α_d are coprime

(iii) Solve the standard Bezout equation

$a(a_l + b_u) + b b_y = a_k a_l$ for b_u, b_y

(iv) Construct 2DOF filters

$$
\begin{aligned}
C_{Ju} &= \frac{b_{Ju}}{a_{Jl}} = \frac{a_{Jl} + b_{Ju} - a_{Jl}}{a_{Jl}} \\
&= \frac{\alpha_d(a_l + b_u) - (\alpha_d a_l + b l_d)}{a_{Jl}} \\
&= \frac{\alpha_d b_u - b l_d}{a_{Jl}}, \text{ stable} \\
C_{Jy} &= \frac{b_{Jy}}{a_{Jl}} \\
&= \frac{a_k l_d + b_y \alpha_d}{a_{Jl}}, \text{ stable}
\end{aligned}
$$

NB: a_l may not be stable - but this is not a problem.

REDUCED DR DESIGN - Continued II

Individual contributions. Can show

$$\begin{aligned}
\overline{\hat{d}} &= C_{du}\overline{u} + C_{dy}\overline{y} \\
C_{du} &= -\frac{bl_d}{a_{Jl}} \\
C_{dy} &= \frac{al_d}{a_{Jl}}
\end{aligned}$$

and if we write

$$k^T\hat{x} = C_{xu}\overline{u} + C_{xy}\overline{y}$$

can show

$$\begin{aligned}
C_{xy} &= \frac{(a_k - a)l_d + b_y a_d}{a_{Jl}} \\
C_{xu} &= \frac{a_d b_u}{a_{Jl}}
\end{aligned}$$

DISTURBANCE CONTROL Revisited

3DOF Disturbance Control TF Lemma.

$$u_{ss,d} = lim_{s \to 0} \frac{s}{\alpha_d(s)}$$

Proof: From earlier results

$$\begin{aligned}
& \lim_{s \to 0} sG_{d \to u}\overline{d}(s) \\
= \quad & \lim_{s \to 0} s \frac{b(l_d a_k + b_y \alpha_d)}{a_{Jl} a_k \alpha_d} \\
= \quad & \lim_{s \to 0} \left(\frac{sbl_d a_k}{a_{Jl} a_k \alpha_d} + \frac{sbb_y}{a_{Jl} a_k}\right) \\
= \quad & \lim_{s \to 0} \frac{sbl_d}{a_{Jl} \alpha_d} + 0 \\
= \quad & \lim_{s \to 0} \frac{s(a_{Jl} - \alpha_d a_l)}{a_{Jl} \alpha_d} \\
= \quad & \lim_{s \to 0} \left(\frac{s}{\alpha_d} - \frac{s a_l}{a_{Jl}}\right) \\
= \quad & \lim_{s \to 0} \frac{s}{\alpha_d}, \quad \text{as claimed.}
\end{aligned}$$

This lemma now makes very clear our earlier result that a step or step+sinusoid can be handled but not a ramp.

DR Parameter Counts

- Control Bezout equation

$a_k a_l = a(a_l + b_u) + bb_y$

same as before

- Joint Observer Polynomial

$a_{Jl} = \alpha_d a_l + bl_d$

Check highest powers

monic(n_{Jl})=monic(n_d) $\times$ monic (n_a)+ lower degree terms

$\Rightarrow n_{Jl} = n_d + n_a$

Check # equations and unknowns

$(n_d + n_a)$ equations $\leftrightarrow \#a_l + \#l_d$

$= n_a$(monic (n_a))

$+n_d$(degree $(n_d - 1)$)

which match.

3DOF DR Example

$$P(s) = \frac{1}{s+3} = \frac{b}{a}, \overline{d} = \frac{1}{s} = \frac{\beta_d}{\alpha_d}$$

Choose: closed loop control pole at -5

observer poles at -10, -10

So $n_a = 1, n_d = 1 \Rightarrow dega_l = n_a = 1,$

$degl_d = n_d - 1 = 0 \Rightarrow a_l = s + \theta, l_d = \gamma$= const.

Then $a_{Jl} = (s+10)^2 = s^2 + 20s + 100$

3DOF DR Example - Continued

(i) Solve $a_{Jl} = \alpha_d a_l + b l_d$

$$\begin{aligned}
\Rightarrow \quad & s^2 + 20s + 100 = s(s+\theta) + 1 \times l_d \\
= \quad & s^2 + \theta s + l_d \\
\Rightarrow \theta \quad = \quad & 20 \Rightarrow a_l(s) = s + 20 \\
l_d \quad = \quad & 100 = \gamma
\end{aligned}$$

(ii) Solve $a_k a_l = a(a_l + b_u) + b b_y$ So
$n_a = 1 \Rightarrow b_u = const., b_y = const.$ So

$$\begin{aligned}
& (s+5)(s+20) = (s+3)(s+20+b_u) + b_y \\
\Rightarrow \quad & 2(s+20) = (s+3)b_u + b_y \\
\Rightarrow \quad & 2 = b_u \\
40 \quad = \quad & 3b_u + b_y \Rightarrow b_y = 34
\end{aligned}$$

3DOF DR Example - Continued II

(iii) C polynomials

$$
\begin{aligned}
C_{Ju} &= \frac{\alpha_d b_u - b l_d}{a_{Jl}} \\
&= \frac{s \times 2 - 1 \times 100}{(s+10)^2} = \frac{2(s-50)}{s^2+20s+100} \\
C_{Jy} &= \frac{a_k l_d + b_y \alpha_d}{a_{Jl}} \\
&= \frac{(s+5)100 + 34s}{(s+10)^2} = \frac{134s+500}{s^2+20s+100}
\end{aligned}
$$

NB:

$$
\begin{aligned}
G_{d\rightarrow y} &= \frac{b\alpha_d(a_l + b_u)}{a_k a_{Jl}} \\
&= \frac{1 \times s \times (s+20+2)}{(s+5)(s+10)^2} \\
&= \frac{s(s+22)}{(s+5)(s+10)^2}
\end{aligned}
$$

and has a factor s as required.

APPENDIX 7A: DERIVATIONS I

$$
\begin{aligned}
a_{Jl} &= \left|sI - A_J + l_J c_J^T\right| \\
&= det \begin{pmatrix} sI - A_d & l_d c^T \\ -bc_d^T & sI - A + lc^T \end{pmatrix} \\
&= |sI - A_d| \\
\times & \left|(sI - A + lc^T) + bc_d^T (sI - A_d)^{-1} l_d c^T\right| \\
&= \alpha_d \left| sI - A + lc^T + bc^T \frac{l_d}{\alpha_d} \right|
\end{aligned}
$$

where we define $\frac{l_d}{\alpha_d} = c_d^T (sI - A_d)^{-1} l_d$

$$
\begin{aligned}
&= \alpha_d \left|sI - A + lc^T\right| \\
\times & \left| I + (sI - A + lc^T)^{-1} bc^T \frac{l_d}{\alpha_d} \right| \\
&= \alpha_d a_l (1 + c^T (sI - A + lc^T)^{-1} b \frac{l_d}{\alpha_d}) \\
\Rightarrow a_{Jl} &= \alpha_d a_l (1 + \frac{b}{a_l} \frac{l_d}{\alpha_d})
\end{aligned}
$$

APPENDIX 7A DERIVATIONS II

since

$$
\begin{aligned}
& c^T(sI - A + lc^T)^{-1}b \\
= \quad & c^T\left[(sI-A)^{-1} - \frac{(sI-A)^{-1}lc^T(sI-A)^{-1})}{1+c^T(sI-A)^{-1}l}\right]b \\
= \quad & c^T(sI-A)^{-1}b\left\{1 - \frac{c^T(sI-A)^{-1}l}{1+c^T(sI-A^{-1})l}\right\} \\
= \quad & \frac{b}{a}(1 - \frac{\frac{l}{a}}{1+\frac{l}{a}}) \\
= \quad & \frac{b}{a}\frac{1}{1+\frac{l}{a}} = \frac{b}{a+l} = \frac{b}{a_l}
\end{aligned}
$$

Continuing $\Rightarrow a_{Jl} = \alpha_d a_l + b l_d$

APPENDIX 7A DERIVATIONS III

$$
\begin{aligned}
& CLTF = c_J^T(sI - A_J + b_J k_J^T)^{-1} b_J \\
= \quad & (0^T, c^T)\left(\begin{bmatrix} sI - A_d & 0 \\ -bc_d^T & sI - A \end{bmatrix} \right. \\
+ \quad & \left. \begin{pmatrix} 0 \\ b \end{pmatrix} \begin{pmatrix} c_d^T, k^T \end{pmatrix} \right)^{-1} \begin{pmatrix} 0 \\ b \end{pmatrix} \\
= \quad & \begin{pmatrix} 0^T & c^T \end{pmatrix} \begin{pmatrix} sI - A_d & 0 \\ 0 & sI - A + bk^T \end{pmatrix}^{-1} \\
\times \quad & \begin{pmatrix} 0 \\ b \end{pmatrix} \\
= \quad & c^T(sI - A + bk^T)^{-1} b \\
= \quad & \frac{b}{a_k} \\
\Rightarrow \quad & a_{Jk} = a_k
\end{aligned}
$$

APPENDIX 7A DERIVATIONS IV

$$\frac{b_{Jy}}{a_J} = k_J^T(sI - A_J)^{-1} l_J$$

$$= \quad (c_d^T, k^T) M \begin{pmatrix} l_d \\ l \end{pmatrix}$$

$$M = \begin{pmatrix} sI - A_d & 0 \\ -bc_d^T & sI - A \end{pmatrix}^{-1}$$

$$= \quad \begin{pmatrix} (sI - A_d)^{-1} & 0 \\ (sI - A)^{-1} bc_d^T (sI - A_d)^{-1} & (sI - A)^{-1} \end{pmatrix}$$

$$\Rightarrow \quad \frac{b_{Jy}}{a_J} = [c_d^T (sI - A_d)^{-1}$$

$$+ \quad k^T (sI - A)^{-1} bc_d^T (sI - A_d)^{-1}, k^T (sI - A)^{-1}] \begin{bmatrix} l_d \\ l \end{bmatrix}$$

$$= \quad c_d^T (sI - A_d)^{-1} l_d (1 + k^T (sI - A)^{-1} b) + k^T (sI - A)^{-1} l$$

$$= \quad \frac{l_d}{\alpha_d}(1 + \frac{\overline{k}}{a}) + \frac{b_y}{a} = \frac{l_d}{\alpha_d}\frac{a_k}{a} + \frac{b_y}{a}$$

$$= \quad \frac{l_d a_k + b_y \alpha_d}{\alpha_d a}$$

as required.

APPENDIX 7A Derivations V

- Claim

$$b_{Ju} + a_{Jl} = \left| sI - A_J + l_J c_J^T + b_J k_J^T \right|$$

- Proof:

$$\begin{aligned} RHS &= \left| sI - A_J + l_J c_J^T \right| \\ &\times \left| I + (sI - A_J + l_J c_J^T)^{-1} b_J k_J^T \right| \\ &= \left| sI - A_J + l_J c_J^T \right| \\ &\times (1 + k_J^T (sI - A_J + l_J c_J^T)^{-1} b_J) \\ &= a_{Jl}(1 + \frac{b_{Ju}}{a_{Jl}}) \\ &= a_{Jl} + b_{Ju} \end{aligned}$$

NB Similarly $b_u + a_l = \left| sI - A + lc^T + bk^T \right|$

APPENDIX 7A Derivations VI

•Now we find also

$$
\begin{aligned}
&RHS = \\
&\left| \begin{pmatrix} sI - A_d & 0 \\ -bc_d^T & sI - A \end{pmatrix} + \begin{pmatrix} l_d \\ l \end{pmatrix} (0^T, c^T) + \begin{pmatrix} 0 \\ b \end{pmatrix} (c_d^T, k^T) \right| \\
&= \\
&\left| \begin{pmatrix} sI - A_d & 0 \\ -bc_d^T & sI - A \end{pmatrix} + \begin{pmatrix} 0 & l_d c^T \\ 0 & lc^T \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ bc_d^T & bk^T \end{pmatrix} \right| \\
&= \begin{vmatrix} sI - A_d & l_d c^T \\ 0 & sI - A + lc^T + bk^T \end{vmatrix} \\
&= |sI - A_d||sI - A + lc^T + bk^T| \\
&= \alpha_d(a_l + b_u)
\end{aligned}
$$

as required.

APPENDIX 7B: FEEDBACK DECOMPOSITIONS

We need to calculate $\overline{\hat{d}} = (c_d^T, 0^T)\overline{\hat{x}}$

Recall

$$\begin{aligned}
\dot{\hat{x}}_J &= A_J\hat{x}_J + b_J u + l_J(y - c_J^T\hat{x}_J) \\
&= (A_J - l_J c_J^T)\hat{x}_J + b_J u + l_J y \\
\Rightarrow \overline{\hat{x}}_J &= (sI - A_J + l_J c_J^T)^{-1} b_J \overline{u} \\
&+ (sI - A_J + l_J c_J^T)^{-1} l_J \overline{y} \qquad (0.1)
\end{aligned}$$

Now in partitioned form

$$sI - A_J + l_J c_J^T = \begin{pmatrix} sI - A_d & l_d c^T \\ -b c_d^T & sI - A + l c^T \end{pmatrix}$$

APPENDIX 7B: FEEDBACK DECOMPOSITIONS II

So by partitioned inversion

$$\begin{aligned}
& (sI - A_J + l_J c_J^T)^{-1} \\
= & \left[\begin{array}{cc} \Delta_d^{-1} & -\Delta_d^{-1} l_d c^T ()_l^{-1} \\ ()_l^{-1} b c_d^T \Delta_d^{-1} & W_l \end{array}\right] \quad (0.2) \\
()_l = & \; sI - A + l c^T \\
\Delta_d = & \; sI - A_d + l_d c^T ()_l^{-1} b c_d^T \\
W_l = & \; ()_l^{-1} - ()_l^{-1} b c_d^T \Delta_d^{-1} l_d c^T ()_l^{-1}
\end{aligned}$$

Now we find

$$\begin{aligned}
\Delta_d &= sI - A_d + l_d \frac{b}{a_l} c_d^T \\
\Rightarrow \Delta_d^{-1} &= ()_d^{-1} - \frac{()_d^{-1} l_d c_d^T ()_d^{-1}}{\frac{a_l}{b} + c_d^T ()_d^{-1} l_d} \\
()_d &= sI - A_d
\end{aligned}$$

However

$$c_d^T ()_d^{-1} l_d = \frac{l_d}{\alpha_d} \quad (0.3)$$

APPENDIX 7B: FEEDBACK DECOMPOSITIONS III

Continuing

$$\Delta_d^{-1} = ()_d^{-1} - \frac{()_d^{-1} l_d c_d^T ()_d^{-1}}{\frac{l_d}{\alpha_d} + \frac{a_l}{b}} \qquad (0.4)$$

Returning to $\bar{d}\hat{}$ we have,

$$\begin{aligned} & \bar{\hat{d}} = (c_d^T, 0^T)\bar{\hat{x}}_J \\ = \quad & [c_d^T \Delta_d^{-1}, -c_d^T \Delta_d^{-1} l_d c^T ()_l^{-1}] \\ \times \quad & \left\{ \begin{pmatrix} 0 \\ b \end{pmatrix} \overline{u} + \begin{pmatrix} l_d \\ l \end{pmatrix} \overline{y} \right\} \end{aligned} \qquad (0.5)$$

APPENDIX 7B: FEEDBACK DECOMPOSITIONS IV

To continue we first calculate that

$$
\begin{aligned}
c_d^T \Delta_d^{-1} &= c_d^T ()_d^{-1} \left[1 - \frac{c_d^T ()_d^{-1} l_d}{\frac{l_d}{\alpha_d} + \frac{a_l}{b}} \right] \\
&= c_d^T ()_d^{-1} (1 - \frac{\frac{l_d}{\alpha_d}}{\frac{l_d}{\alpha_d} + \frac{a_l}{b}}), \\
&= c_d^T ()_d^{-1} \frac{\frac{a_l}{b}}{\frac{l_d}{\alpha_d} + \frac{a_l}{b}} \\
&= c_d^T ()_a^{-1} \frac{a_l \alpha_d}{b l_d + \alpha_d a_l} \\
&= c_d^T ()_d^{-1} \frac{a_l \alpha_d}{a_{Jl}} \qquad (0.7)
\end{aligned}
$$

We now find that

$$
\begin{aligned}
\overline{\hat{d}} &= C_{du} \overline{u} + C_{dy} \overline{y} \\
C_{du} &= -c_d^T \Delta_d^{-1} l_d c^T ()_l^{-1} b \\
C_{dy} &= c_d^T \Delta_d^{-1} l_d - c_d^T \Delta_d^{-1} l_d c^T ()_l^{-1} l
\end{aligned}
$$

APPENDIX 7B: FEEDBACK DECOMPOSITIONS V

We find

$$
\begin{aligned}
c_d^T \Delta_d^{-1} l_d &= c_d^T ()_d^{-1} l_d \frac{a_l \alpha_d}{a_{Jl}} \\
&= \frac{l_d a_l}{a_{Jl}}
\end{aligned}
$$

Thus since $c^T ()_l^{-1} b = \frac{b}{a_l}$ we get

$$
C_{du} = -\frac{l_d a_l}{a_{Jl}} \frac{b}{a_l} = -\frac{b l_d}{a_{Jl}} \tag{0.8}
$$

While, since $c^T ()_l^{-1} l = \frac{l}{a_l}$

$$
\begin{aligned}
C_{dy} &= \frac{l_d a_l}{a_{Jl}}(1 - \frac{l}{a_l}) = \frac{l_d a_l}{a_{Jl}} \frac{a}{a_l} \\
&= \frac{l_d a}{a_{Jl}} \qquad (0.9)
\end{aligned}
$$

APPENDIX 7B: FEEDBACK DECOMPOSITIONS VI

We get C_{xu}, C_{xy} by subtraction

$$
\begin{aligned}
C_{xu} &= C_{Ju} - C_{du} \\
&= \frac{\alpha_d b_u - b l_d}{a_{Jl}} + \frac{b l_d}{a_{Jl}} = \frac{\alpha_d b_u}{a_{Jl}} \\
C_{xy} &= C_{Jy} - C_{dy} \\
&= \frac{a_k l_d + b_y \alpha_d}{a_{Jl}} - \frac{a l_d}{a_{Jl}} \\
&= \frac{(a_k - a) l_d + b_y \alpha_d}{a_{Jl}}
\end{aligned}
$$